

Dear Ms. Wallenwein,

I'm applying for the AI Engineer - Transfer Pricing position. I work at the intersection of applied AI research and software engineering, and building systems that turn expert knowledge into something scalable is exactly the kind of problem I want to keep working on.

My Master's thesis at RPTU Kaiserslautern tackles this directly: a reinforcement learning system that repairs logical inconsistencies in OWL 2 DL knowledge graphs, combining a Double DQN agent with formal ontology reasoning. I integrated GPT-4o mini for triple extraction and fuzzy ontology alignment, and validated the pipeline on a knowledge graph with over 2,000 triples, reaching 88-96% repair success. It's the same combination this role asks for - LLMs and knowledge graphs working together in a real pipeline, not just a notebook experiment.

At Fraunhofer IGCV I worked on industrial AI systems end to end: building knowledge graphs in Neo4j, combining causal discovery with graph neural networks for root cause analysis, and deploying an anomaly detection model that reached about 90% accuracy in production. I also built the dashboards and APIs that domain experts without an AI background used directly, which meant translating model output into something they could act on without me in the room.

That's also true of my current work on TUM's Koinon platform, where I build features for university administrators, and of my time at DFKI, where I built a debugging tool for researchers working on multi-agent systems. Explaining technical work clearly to people who aren't technical is something I do by default, not something I'm still learning.

I'd welcome the chance to bring this mix of AI and full-stack engineering to your Transfer Pricing team, and I'm genuinely interested in how international tax questions get restructured into something a model can help with. I'm based in Kaiserslautern, available immediately, fluent in English, and at B1 level in German.

Best regards,

Resham Joshi